A Statistical Study on the Automobile Sector In India

A PROJECT REPORT

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CERTIFICATE

This is to certify that **Ms. Saroj Karwal** , a student of M.Sc. Statistics IV Semester, Department of Statistics, University of Rajasthan, has successfully prepared and submitted this project report entitled: **"A Statistical Study on the Automobile Sector in India"** .The report is based on secondary data and has been completed under my supervision and guidance as part of the curriculum requirement for the M.Sc. IV Semester program.

This project is being submitted in partial fulfilment of the requirements for the Master of Science degree in Statistics at the University of Rajasthan.

I further certify that this project is the original work of the student and has been completed with academic integrity. The report has not been copied from any other source and reflects the student's own understanding and analysis.

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SAROJ KARWAL

PREFACE

This study explores the trends and patterns in the production, sales, and export data of the automobile sector in India. The project begins with the use of Descriptive Statistics to summarize and interpret the core characteristics of the data. Measures such as mean, median, standard deviation, and range help in understanding the distribution and variability in automobile production, domestic sales, and export volumes over time.

Building on this foundation, the study employs Time Series Analysis to examine the temporal behaviour of the data. Techniques such as trend analysis, seasonal decomposition, and forecasting models are utilized to identify patterns, cyclical movements, and potential future directions in the sector's performance.

By combining descriptive and time-based analytical methods, this study provides a comprehensive view of the automobile industry's operational dynamics. The insights gained can support strategic planning, policy development, and market analysis in one of the most dynamic sectors of the global economy.

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1. INTRODUCTION:-

The automobile sector plays a pivotal role in the global economy and is one of the most influential industries in terms of employment, technological advancement, and environmental impact. As a major contributor to a country's GDP, the sector drives growth in manufacturing, trade, and services. In India, the automobile industry stands as one of the largest in the world, offering a wide range of vehicles including two-wheelers, passenger cars, commercial vehicles, and electric vehicles (EVs).

Globally, the automobile industry has evolved significantly over the last century. From the inception of mass-produced cars by Ford in the early 20th century to the rise of electric and autonomous vehicles today, the sector has continually adapted to consumer demand, technological advancements, and shifting regulatory frameworks. The introduction of electric vehicles, in particular, has marked a new era, with countries setting ambitious targets for zero-emission vehicles to combat climate change.

In India, the automobile industry is a key contributor to economic growth. It not only provides employment to millions but also supports a vast network of ancillary industries, including parts manufacturing, supply chains, and logistics. As of recent years, India has emerged as a major hub for automobile manufacturing, with global automotive giants such as Maruti Suzuki, Hyundai, and Tata Motors having significant operations in the country.

India is the world’s largest manufacturer of two-wheelers and is steadily increasing its share of passenger car and commercial vehicle production. Additionally, with a growing middle class and

increased urbanization, the demand for personal vehicles continues to rise, providing further opportunities for growth.

This study aims to examine the performance of the automobile sector through statistical analysis. It will explore key indicators such as vehicle production, sales trends, export patterns, and the shift towards electric vehicles. The study also evaluates factors affecting production, sales, and export, such as economic conditions, government policies, and global supply chain issues. Furthermore, the economic and environmental impact of the industry will be assessed, providing valuable insights into the sector's future trajectory.

By applying statistical tools, such as regression analysis, trend analysis, and forecasting models, this study will provide a clear picture of the sector’s current state and future potential.

2.TYPES OF VEHICLES IN THE AUTOMOBILE SECTOR:-

The automobile sector includes a wide variety of vehicles that are designed to serve different purposes. These vehicles can be classified into several types based on their use, structure, number of wheels, fuel type, and design. This classification helps in understanding the diversity and functionality of the vehicles in the industry.

1.1 Classification Based on Usage:-

a. Passenger Vehicles:-

These vehicles are primarily used for carrying people. They are designed to offer comfort and convenience for individuals or families. Common types of passenger vehicles include hatchbacks, sedans, SUVs (Sports Utility Vehicles), and MUVs (Multi Utility Vehicles). They are often used for daily commuting, travel, and personal transportation.

b. Commercial Vehicles:-

Commercial vehicles are designed for transporting goods or people for business purposes. These include trucks, buses, vans, and pickups. They play a major role in logistics, supply chains, construction, and public transport systems. Depending on their capacity and size, they are further divided into light, medium, and heavy commercial vehicles

1.2 Classification Based on Number of Wheels

Vehicles are classified by the number of wheels to differentiate their capacity, balance, and usage. This classification helps in regulatory laws, licensing, insurance, and also understanding their application in transport systems.

a. Two-Wheelers

These are single-track vehicles with two wheels, generally powered by internal combustion engines or electric motors. They are lightweight, easy to maneuver, and have low running costs, making them ideal for urban commuting. Motorcycles (Royal Enfield, Pulsar), Scooters (Honda Activa , TVS Jupiter), Mopeds.

Applications: Personal use, Delivery services, Rural and semi-urban commuting

b. Three-Wheelers

These are intermediate vehicles between two-wheelers and four-wheelers. They provide more stability and load capacity than two-wheelers while remaining economical. Typically powered by petrol, diesel, CNG, or even electric batteries. Auto-rickshaws, Cargo vehicles like Piaggio Ape.

Application: Public transport, Goods delivery, Shared ride services (Ola auto, Rapido)

c. Four-Wheelers and Multi-Axle Vehicles

Four-wheelers are the most common category used for personal and commercial purposes. Multi-axle vehicles have additional axles to support higher loads, used for logistics, construction, and mass transport. Cars, SUVs (Tata Nexon, Hyundai Creta), Trucks (Ashok Leyland, BharatBenz), Buses (Volvo, Tata Starbus)

Applications: Personal and family transport, Long-distance goods movement, Public and private bus services

1.3 Classification Based on Fuel Type

This classification is based on the energy source used for engine operation. It's vital for studying pollution impact, fuel efficiency, environmental regulations, and government policies like emission norms (e.g., BS6, Euro VI).

a. Petrol Vehicles

These vehicles use a spark-ignition engine that burns petrol to generate power. They offer smooth acceleration, are lighter in design, but are less fuel-efficient than diesel. Cars, bikes used for personal use or short commutes.

b. Diesel Vehicles

Diesel vehicles use compression-ignition engines. They provide better mileage and torque, making them ideal for heavy vehicles and long-distance travel. Buses, trucks, some SUVs.

c. Electric Vehicles (EVs)

EVs are powered by electric motors using lithium-ion batteries. They produce zero tailpipe emissions, aligning with sustainable development goals. Low running cost, Supported by subsidies under schemes like FAME-II, Tata Tiago EV, Ola S1, MG ZS EV

d. Hybrid Vehicles

These vehicles combine a traditional engine (petrol/diesel) with an electric motor. The goal is to optimize fuel use and reduce emissions. Toyota Camry Hybrid, Honda City e:HEV

e. CNG/LPG Vehicles

CNG (Compressed Natural Gas) and LPG (Liquefied Petroleum Gas) are clean-burning fuels. Vehicles using these fuels emit fewer pollutants and are economical for commercial use. Auto-rickshaws, city buses, taxis

f. Hydrogen Fuel Cell Vehicles

These use electrochemical cells to convert hydrogen gas into electricity. They emit only water vapor, offering a zero-emission solution for the future. Still emerging due to infrastructure limitations. Toyota Mirai, Hyundai Nexo

1.4 Classification Based on Body Design

Body design influences a vehicle's aerodynamics, interior space, utility, and target users. Automakers use this classification to cater to diverse consumer needs, from city cars to off-roaders and luxury models.

a. Hatchbacks

Compact cars with a rear hatch-type door, offering flexible storage. Known for affordability and city drivability. Maruti Swift, Hyundai i20

b. Sedans

Larger cars with a three-box design (engine, cabin, boot). Offer more comfort and premium features. Examples: Honda City, Skoda Slavia

c. SUVs (Sports Utility Vehicles)

High-ground clearance, often equipped with 4WD/AWD systems. Suitable for both urban and rough terrain. Examples: Mahindra Scorpio, Tata Harrier

d. MUVs (Multi Utility Vehicles)

Built to carry multiple passengers and cargo. These vehicles prioritize space and utility.

Toyota Innova, Maruti Ertiga

e. Convertibles and Coupes

Convertibles: Roof can be folded or removed, typically luxury or sports cars. Coupes: Sleek two-door cars with sporty looks and performance.

BMW Z4 (Convertible), Audi TT (Coupe)

f. Pick-up Trucks and Vans

Pick-up Trucks: Open rear bed for carrying goods. Vans: Enclosed structure to carry people or cargo. Tata Yodha (Pickup), Force Traveller (Van)

3 KEY INDICATORS AND SECTOR OVERVIEW:-

The automobile sector is a critical component of a nation's industrial and economic framework. It reflects the level of technological advancement, infrastructure development, and economic maturity of a country. To study and evaluate the performance of this sector, economists and researchers rely on certain key indicators, which offer insights into the sector's growth, contribution, and transformation over time.

3.1 Production performance:-

Production is one of the most fundamental indicators of the automobile industry. It signifies the manufacturing output of vehicles over a defined time period. This indicator reflects the sector's capacity utilization, supply-side efficiency, and ability to meet domestic as well as international demand.

3.2 Sales and Market Demand:-

Sales data, both in domestic and international markets, serve as an indicator of consumer demand, market trends, and purchasing power. Steady or rising sales figures are generally considered a sign of sectoral health, economic stability, and product acceptance among consumers.

3.3 Employ Contribution:-

The automobile sector is labor-intensive and provides substantial employment opportunities, both directly and indirectly. It includes jobs in vehicle production, sales, after-sales services, spare parts manufacturing, logistics, and marketing. Thus, employment generation is an important social and economic indicator of the sector.

3.4 Investment and Capital Formation:-

Capital investment in the automobile sector indicates industrial growth and technological advancement. Both domestic and foreign direct investment (FDI) are essential for setting up manufacturing plants, research and development centers, and supply chain systems. The flow of investment reflects business confidence and policy effectiveness.

3.5 Technological Innovation and Fuel Diversification:-

The adoption of new technologies, including electric mobility, hybrid engines, safety systems, and automation, is a key indicator of the sector's modernization. A shift towards alternative fuels and cleaner technologies represents a commitment to sustainable development and environmental protection.

3.6 Infrastructure Support and Policy Framework:-

The sector's growth is closely tied to infrastructure development such as road networks, charging stations, and fuel availability. Moreover, government policies and incentives also serve as indicators of institutional support and sectoral direction.

3.7 Sector Overview:-

The automobile industry encompasses a wide range of vehicle categories including two-wheelers, three-wheelers, passenger cars, and commercial vehicles. Each segment serves a unique economic and social function—ranging from personal mobility to large-scale freight movement. The sector not only supports industrial output but also stimulates demand in ancillary industries like steel, rubber, glass, plastic, electronics, and petroleum.

Furthermore, the automobile sector acts as a catalyst for technological progress and innovation. With the global focus shifting toward sustainable and green mobility solutions, the industry is undergoing a phase of transformation, marked by increasing investment in electric vehicles, hybrid technologies, and smart transportation systems.

The strategic importance of the automobile sector lies in its capacity to generate employment, support trade, contribute to GDP, and drive industrial diversification. It serves as a barometer of economic progress and an engine of growth for developing as well as developed economies.

In essence, the key indicators of the automobile sector—ranging from production and sales to innovation and employment—provide a comprehensive view of its overall performance and future potential. A theoretical understanding of these indicators allows policymakers, researchers, and industry stakeholders to make informed decisions that promote sustainable and inclusive growth within the sector.

4 FACTORS AFFECTING PRODUCTION, SALES AND EXPORT:-

The automobile sector is influenced by a wide range of factors that impact its production output, market sales, and export performance. These factors can be broadly categorized into economic, technological, policy-related, environmental, and global aspects. Understanding these factors is essential for analyzing the dynamics of the industry and for formulating strategies to sustain growth.

4.1 Factors Affecting Production:-

1. Availability of Raw Materials

The automobile manufacturing process relies heavily on materials like steel, aluminum, rubber, plastics, and electronic components. Any fluctuation in the availability, cost, or import of these materials can directly influence the production process.

2. Labor and Workforce Efficiency

The quality and availability of skilled labor are critical for efficient production. Strikes, labor shortages, or lack of technical training may hinder production timelines.

3. Technological Infrastructure

Access to modern technology, automated machinery, and robotics significantly enhances production capacity and quality. Lack of technological advancement can limit production efficiency and competitiveness.

4. Supply Chain and Logistics

A smooth supply chain ensures timely delivery of parts and materials. Disruptions due to transportation delays, natural disasters, or geopolitical tensions can affect production schedules.

5. Government Policies and Industrial Regulations

Policies related to industrial licensing, environmental clearances, tax incentives, and labor laws affect the ease and cost of production. Supportive policies encourage higher output, while stringent regulations may slow it down.

4.2 Factors Affecting Sales:-

1. Consumer Demand and Preferences

The demand for automobiles is influenced by consumer income levels, lifestyle, preferences, and brand perception. Urbanization and changing family structures also shape buying patterns.

2. Price and Affordability

The cost of vehicles, availability of finance, and interest rates play a crucial role in consumer decisions. Price-sensitive markets may react sharply to even small changes in prices.

3. Marketing and Distribution Networks

Effective marketing strategies and wide dealership networks help in increasing customer reach. Brand awareness and after-sales service also influence purchasing decisions.

4. Fuel Prices and Maintenance Costs

Rising fuel costs or high maintenance expenses can reduce the appeal of certain vehicle segments. Consumers often consider long-term affordability in their choices.

5. Seasonal and Festive Trends

In many countries, including India, automobile sales rise during festivals and harvest seasons due to cultural factors and increased disposable income.

4.3 Factors Affecting Export:-

1. International Trade Policies

Tariffs, import-export duties, and trade agreements between countries greatly affect the cost and feasibility of vehicle exports. Favorable policies promote exports, while protectionist measures may restrict them.

2. Global Market Demand

The demand for automobiles in foreign markets depends on income levels, road infrastructure, fuel availability, and regulatory standards. Different countries prefer different vehicle types and features.

3. Currency Exchange Rates

A stable and favorable exchange rate encourages exports by making domestic products more affordable in foreign markets. Unstable currencies can lead to losses for exporters.

4. Quality Standards and Certifications

International markets require compliance with strict safety and emission norms. Only vehicles meeting these standards can be exported, which requires manufacturers to invest in advanced technology.

5. Shipping and Logistics Infrastructure

Efficient ports, shipping routes, and transportation systems are necessary for timely and cost-effective exports. Any disruption in the logistics chain can lead to delays and increased costs.

The production, sales, and export of vehicles are influenced by a combination of internal and external factors. While efficient production depends on raw materials, labor, and infrastructure, sales are driven by consumer behavior, pricing, and economic conditions. Export performance, on the other hand, is largely shaped by global market dynamics and trade regulations. A holistic understanding of these factors helps policymakers and industry leaders make informed decisions to support sustainable growth in the automobile sector.

5 ECONOMIC AND ENVIORMENTAL IMPACT:-

The automobile sector exerts a profound influence on both the economic and environmental landscapes of nations. As a driver of industrialization and urbanization, it contributes significantly to national economic development. Simultaneously, however, it imposes substantial ecological burdens, necessitating a balanced approach to policy and innovation.

5.1 Economic Impact:-

1. Industrial Backbone and GDP Contribution

The automobile industry forms a critical component of a country's industrial base, contributing a substantial share to the Gross Domestic Product (GDP). It acts as a multiplier sector—stimulating demand across a spectrum of upstream and downstream industries including metallurgy, petrochemicals, electronics, finance, and logistics.

2. Employment and Skill Development

Beyond direct employment in assembly lines and R&D labs, the sector supports a vast ecosystem of indirect jobs, including parts manufacturing, vehicle maintenance, retail sales, insurance, and transportation. Furthermore, it fosters the development of skilled labor through vocational training and technical education, contributing to human capital formation.

3. Capital Formation and Foreign Investment

As a capital-intensive sector, it attracts both domestic investment and Foreign Direct Investment (FDI), particularly in emerging markets. These investments enhance technological capabilities, support capacity expansion, and integrate the domestic industry into global value chains (GVCs).

4. Infrastructure and Urban Growth Catalyst

The expansion of the automobile industry directly influences the development of complementary infrastructure such as roadways, bridges, expressways, service stations, and parking facilities. This, in turn, accelerates urbanization and spatial economic development, especially in peri-urban and semi-urban regions.

5. Trade Competitiveness and Export Orientation

Countries with mature automobile manufacturing capabilities often position themselves as key exporters of vehicles and auto components. A robust automotive export base contributes positively to the balance of trade and strengthens a nation's industrial competitiveness in the global market.

5.2 Environmental Impact

1. Emissions and Atmospheric Degradation

The automobile sector is a major source of anthropogenic emissions, including carbon dioxide (CO_2), methane (CH_4), nitrogen oxides (NO_x), and volatile organic compounds (VOCs). These emissions exacerbate urban air pollution, acid rain, and smog formation, posing serious risks to public health and environmental stability.

2. Contribution to Climate Change

Internal combustion engine (ICE) vehicles are among the largest contributors to global greenhouse gas (GHG) emissions. The sector's carbon footprint is a central concern in climate policy, prompting regulatory interventions such as Corporate Average Fuel Economy (CAFE) standards, carbon pricing, and decarbonization targets under global frameworks like the Paris Agreement.

3. Fossil Fuel Dependency and Energy Security

The industry's reliance on petroleum-based fuels undermines energy sustainability and subjects economies to geopolitical risks and oil price volatility. This dependency also impedes progress toward renewable energy integration and energy independence.

4. Acoustic and Thermal Pollution

Beyond air quality, automobiles contribute to urban noise pollution, which affects cognitive and cardiovascular health. In densely populated areas, excessive vehicle density also contributes to localized heat islands, disrupting microclimates.

5. Waste Management and Lifecycle Impacts

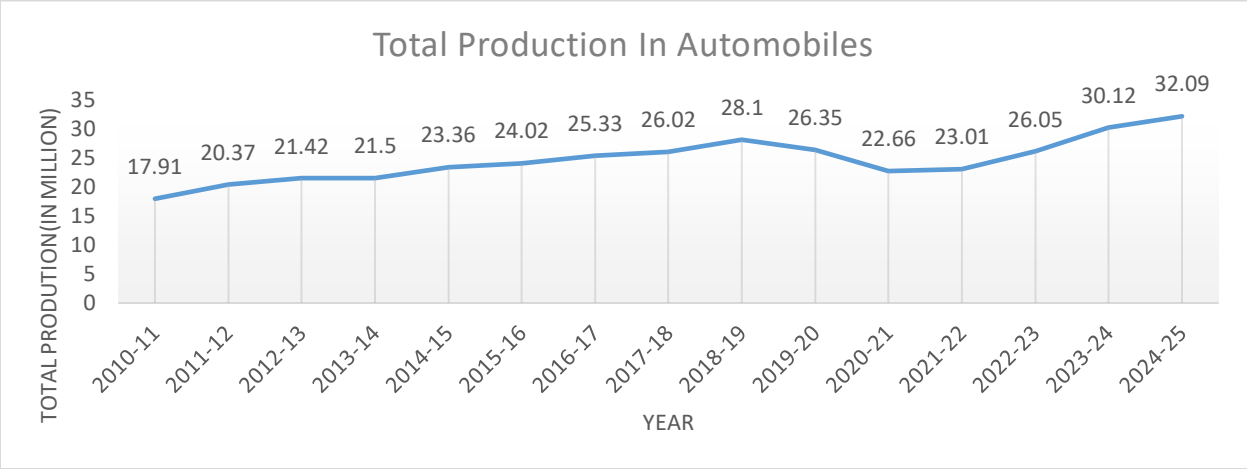
Automobile production and disposal generate significant industrial waste, including heavy metals, plastics, lubricants, and non-biodegradable materials. End-of-life vehicles (ELVs) pose a challenge for recycling systems, especially in the absence of circular economy models and extended producer responsibility (EPR) policies.

The automobile sector stands as a paradoxical pillar—an indispensable engine of economic modernization and simultaneously a substantial ecological stressor. Its dual impact necessitates a strategic shift toward sustainable mobility. This includes investment in green technologies, promotion of electric vehicles (EVs), enhancement of public transportation systems, and the development of a circular economy in vehicle manufacturing and disposal. The future of the sector lies in balancing economic imperatives with environmental stewardship—driving growth without compromising the planet.

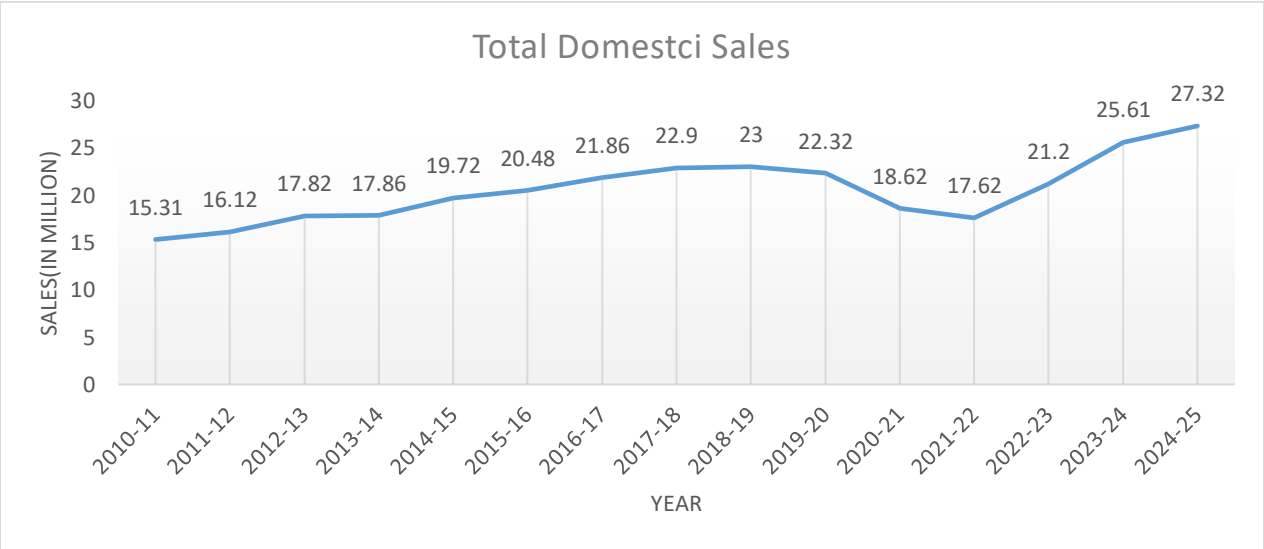
6 DATA PRESENTATION ON PRODUCTION , SALES AND EXPORT:-

The data of production ,total domestic sales and total export used in this study is obtained from the official reports and publications of the **Society of Indian Automobile Manufacturers (SIAM)** since 2010

Year	Total production (In Million)	Total Domestic sales (In Million)	Total Export (In Million)
2010-11	17.91	15.31	2.33
2011-12	20.37	17.38	2.91
2012-13	21.42	17.82	2.90
2013-14	21.50	17.86	3.09
2014-15	23.36	19.72	3.57
2015-16	24.02	20.48	3.64
2016-17	25.33	21.86	3.58
2017-18	26.02	24.97	4.04
2018-19	28.10	26.27	4.62
2019-20	26.35	21.55	4.74
2020-21	22.66	18.62	4.90
2021-22	23.01	17.62	5.00
2022-23	26.05	21.20	5.26
2023-24	30.12	25.61	5.20
2024-25	32.09	27.32	5.40



Graph 1: Total production in automobile sector



Graph 2: Total domestic sales in automobile sector

Graph 3: Total export in automobile sector

7 STATISTICAL DESCRIPTION AND FORMULAE

7.1 Descriptive Statistics :-

Mean:-

The mean, or arithmetic average, is calculated by summing all the values in a dataset and dividing by the total number of values. It's sensitive to outliers and is commonly used when the data is symmetrically distribute

$$\overline{X} = \frac{\sum x}{n}$$

where $n = \text{Total no. of observastion}$

$\sum x = Total$ of all observation

Median:-

The median is the middle value when the dataset is arranged in ascending or descending order. If there's an even number of values, it's the average of the two middle values. The median is robust to outliers and is often used when the data is skewed.

Median for odd data: $M = \left(\frac{n+1}{2}\right)^{th} observation$

Median for even data : $M = \frac{\left(\frac{n+1}{2}\right)^{th} obs + \left(\frac{n}{2}\right)^{th} obs}{2}$

Skewness:-

Skewness can be defined as a statistical measure that describes the lack of symmetry or asymmetry in the probability distribution of a dataset. It quantifies the degree to which the data deviates from a perfectly symmetrical distribution, such as a normal (bell-shaped) distribution. Skewness is a valuable statistical term because it provides insight into the shape and nature of a dataset's distribution.

$$S_K = \frac{3(\overline{X} - M)}{\sigma}$$

where S_K = Skewness

$\overline{X}$ = Mean

M =Median

σ = Stanadard deviation

Kurtosis:-

Kurtosis is a statistical measure that defines how heavily the tails of a distribution differ from the tails of a normal distribution. In other words, kurtosis identifies whether the tails of a given distribution contain extreme values

$$\text{Kurt}(x)=E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right]$$

Where $\mu=E(X)$

$\Sigma =$ Stanadard deviation

7.2 Correlation:

Correlation coefficient can be defined as a measure of the relationship between two quantitative or qualitative variables, i.e. X and Y. It serves as a statistical tool that helps to analyse and in turn, measure the degree of the linear relationship between the variables.

For example, a change in the monthly income (X) of a person leads to a change in their monthly expenditure (Y). With the help of correlation, you can measure the degree up to which such a change can impact the other variables.

7.2.1 Karl Pearson's correlation coefficient: -

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Karl Pearson's correlation coefficient also known as the Product Moment Correlation Coefficient and was developed by Karl Pearson. It is one of three most potent and extensively used methods to measure the level of correlation.

Formula is given by:

$$r = \frac{cov(x, y)}{\sigma_x \sigma_y} \ldots . (1)$$

Where cov(x,y) is covariance of variable X and Y.

This formula can also be written as:

$$r_{xy} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - \left(\sum x_i\right)^2} \sqrt{n \sum y_i^2 - \left(\sum y_i\right)^2}}.$$

Some properties of Karl Pearson's correlation coefficient are:

i. The Correlation Coefficient (r) does not have any unit.
ii. r with a positive value signifies that both X and Y move along the same direction.
iii. r with a negative value indicates an inverse relation between X and Y.
iv. X and Y are said to be not correlated if the value of r is 0.
v. r with a high value signifies a strong linear relationship between two variables.
vi. r with a low value signifies a weak relationship between two variables.
vii. Correlation between two variables is said to be perfect if the value of r is either +1 or –

7.3 Time series analysis: -

When quantitative data are arranged in the order of their occurrence, the resulting statistical series is called a time series. The quantitative values are usually recorded over equal time interval daily, weekly, monthly, quarterly, half yearly, yearly, or any other time measure. Monthly statistics of

Industrial Production in India, Annual birth-rate figures for the entire world, yield on ordinary shares, weekly wholesale price of rice, daily records of tea sales or census data are some of the examples of time series. Each has a common characteristic of recording magnitudes that vary with passage of time.

A time series is a sequence of quantitative data recorded at regular time intervals such as daily, monthly, or yearly. Examples include industrial production, birth rates, stock prices, or rice prices. The main feature of a time series is that its values vary over time.

7.3.1 Secular (Long-Term) Trend

The secular trend shows the long-term direction (growth or decline) in a time series, unaffected by seasonal or short-term factors. This could result from factors like population growth, technological progress, or changing preferences.

Trends can be:

- Linear (constant growth),
- Exponential (constant percentage growth),
- Or complex (like S-curves).

Recognizing trends helps in measuring other components like seasonal or cyclical effects.

7.3.2 Seasonal Variations

These are repetitive changes within a year, often due to climate or customs (e.g., crop production, Diwali sales). Businesses plan around these predictable changes, and seasonal variations are measured as percentages of the trend using seasonal indices.

7.3.3 Business Cycles

These are economic fluctuations that show phases like growth, decline, and recovery, occurring due to complex market forces. They are irregular but longer in duration than seasonal changes. Business cycles are measured by comparing actual values with trend and seasonal components.

7.3.4 Irregular (Erratic) Variations

These are caused by unpredictable events like wars, floods, or economic crises. They don't follow any pattern and are hard to measure. Examples include the Great Depression or post-war booms.

Methods of Measuring Trend

1. Freehand Curve Method: Plot the data and draw a smooth line through it manually.
2. Semi-Averages Method: Divide data into two halves, find their averages, and draw a line between them.
3. Moving Average Method: Smooth the data using a sliding window of averages to reveal trends.
4. Least Squares Method: Fit a trend line mathematically such that the sum of squared deviations is minimized—this is the most accurate method.

1. Freehand Curve Method:

The term freehand is used to any non-mathematical curve in statistical analysis even if it is drawn with the aid of drafting instruments. This is the simplest method of studying trend of a time series. The procedure for drawing free hand curve is as follows:

(i) The original data are first plotted on a graph paper.

(ii) The direction of the plotted data is carefully observed.

(iii) A smooth line is drawn through the plotted points.

While fitting a trend line by the freehand method, an attempt should be made that the fitted curve conforms to these conditions.

(i) The curve should be smooth either a straight line or a combination of long gradual curves.

(ii) The trend line or curve should be drawn through the graph of the data in such a way that the areas below and above the trend line are equal to each other.

(iii) The vertical deviations of the data above the trend line must equal to the deviations below the line. Sum of the squares of the vertical deviations of the observations from the trend should be minimum.

2. Trend by the Method of Semi-averages:

This method can be used if a straight-line trend is to be obtained. Since the location of only two points is necessary to obtain a straight-line equation, it is obvious that we may select two representative points and connect them by a straight line. Data are divided into two halves and an average is obtained for each half. Each such average is shown against the mid-point of the half period, we obtain two points on a graph paper. By joining these points, a straight-line trend is obtained. The method is to be commended for its simplicity and used to some extent in practical work. This method is also flexible, for it is permissible to select representative periods to determine the two points. Unrepresentative years may be ignored.

3. Moving average method:

It consists in measurement of trend by smoothing out the fluctuations of the data by means of a moving average of extent m is a series of successive average is the mean of the first 'm' terms; the second is the mean of m terms from second to $(m+1)^{th}$ terms, thirds is the mean of m terms from 3rd to $(m+2)^{th}$ terms and so on.

If m is odd= (2k+1) say, moving average is placed against the mid-value of the time interval it covers, i.e., against t=k+1 and if m is even =2k (say), it is placed t=k+1 and if m is even = 2k (say),

it placed between the two middle value of the time interval it covers, i.e., against t=k+1. In the latter case the moving average does not coincide with an original time period and an attempt is made to synchronise the moving average and the original data by centring the moving averages which consists in taking a moving average of extent two, of this moving average and putting the first of these value against values t=k+1. The graph obtained on plotting the moving average value against the corresponding time value gives trend curve.

4. Method of Least Squares:

If a straight line is fitted to the data it will serve as a satisfactory trend, perhaps the most accurate method of fitting is that of least squares.

This method is designed to accomplish two results.

(i) The sum of the vertical deviations from the straight line must equal zero.

(ii) The sum of the squares of all deviations must be less than the sum of the squares for any other conceivable straight line.

There will be many straight lines which can meet the first condition. Among all different lines, only one line will satisfy the second condition. It is because of this second condition that this method is known as the method of least squares. It may be mentioned that a line fitted to satisfy the second condition, will automatically satisfy the first condition.

8 APPLICATIONS OF STATISTICAL TOOLS

8.1 Descriptive Statistics:-

Here we find descriptive to know the overall overview of total production , total sales and total export of automobile since 2010.

Variables	Mean	Median	Standard devi.	Skewness	Kurtosis
Total Production	24.76	24.02	3.887116	0.168602	-0.70826
Total Domestic Sales	20.67533	20.48	3.347884	0.297286	-0.97001
Total Export	4.113333	4.04	1.048561	-0.14086	-1.36139

8.2 Correlation:-

Here we find that there are any correlation between different variable in automobile sector like total production , total domestic sale and total export since 2010

Variable 1	Variable 2	correlation
Total production	Total Domestic Sales	r=0.9842
Total production	Total Export	r=0.6688
Total Export	Total Domestic Sales	r=0.5433

8.3 Time series analysis:-

Here first we fit curve using free hand curve method. From table 1, we have data for total Production ,total domestic sales and total export.

Graph (1), Graph (2), Graph (3) and represent free hand curve for total production, total domestic sales and total export in automobile.

As we can see that curve fitted by free hand curve method in Graph (1), (2), (3) is almost linear type of curves, since they have downward shift from year 2019 to 2012 due to covid 19 pandemic , so they are counted as irregular flcutaions in graph and can be ignored from graph .so finally our data is linear ,Now we do trend analysis using least square method

So, trend is analysed using least square method, if we assume linear type of relation between total production and years.

If we assume following linear relation between time(t) and total production in automobiles Y_t:

$Y_t = a + b*x$ ……. (A)

formula for trend analysis by least square method we get:

$\Sigma Y_t = n*a + b\Sigma x$ …… (B)

$\Sigma xY_t = a\ \Sigma x + b\Sigma x^2$ …… (C)

We solve above equation to get value of 'a' and 'b' then we get trend equation by putting these values in equation (A). Now solve equation (B) and (C) for total domestic sales and total export in automobile sector to get their trend equation.

8.3.1 Trend Equation for production:-

Taking data from table 1 we find value of 'a' and 'b'. Values of a and b, by solving equation (B), (C) are

a = 18.339

b = 0.911

now trend equation:

Yt=18.479+0.8929*x where x = t-2011

Now trend values table:

year	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18
Actual values	17.91	20.37	21.42	21.50	23.36	24.02	25.33	26.02
Trend values	19.20	20.16	21.07	21.98	22.89	23.80	24.71	25.62

Year	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Actual values	28.10	26.35	22.66	23.01	26.05	30.12	32.09
Trend values	26.53	27.44	28.36	29.27	30.18	31.09	32.00

8.3.2 Trend Equation for domestic sales:-

Taking data from table 1 we find value of 'a' and 'b'. Values of a and b, by solving equation (B), (C) are

a = 15.18

b = 0.8246

now trend equation:

Y_t =15.18 +0.8246 *x where x = t-2011

Now trend values table:

Year	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18
Actual values	15.321	16.12	17.82	17.86	19.72	20.48	21.86	22.90
Trend values	16.00	16.82	17.65	18.47	19.30	20.12	20.95	21.77

Year	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Actual values	23.00	22.32	18.62	17.62	21.20	25.61	27.32
Trend values	22.60	23.42	24.25	25.07	25.89	26.72	27.54

8.3.3 Trend Equation for Export:-

Taking data from table 1 we find value of 'a' and 'b'. Values of a and b, by solving equation (B), (C) are

a = 2.2941

b = 0.2196

now trend equation:

Y_t =2.2941 +0.2196 *x where x = t-2011

Now trend values table:

year	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18
Actual values	2.33	2.91	2.90	3.09	3.57	3.64	3.58	4.04
Trend values	2.51	2.73	2.95	3.17	3.39	3.61	3.83	4.05

Year	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
Actual values	4.20	4.74	4.90	5.00	5.26	5.20	5.40
Trend values	4.27	4.49	4.70	4.92	5.14	5.36	5.58

9 FORECASTING :-

Using above trend equations we find the future values for production , sales and export in automobile sector

i) For total production:-

Trend equation is : Yt=18.479+0.8929*x where x = t-2011

YEAR	2025-26	2026-27	2027-28	2028-29	2029-30
FORECASTED	32.76	33.65	34.55	35.44	36.33

ii) For total domestic sales:-

Trend equation is : Y_t =15.18 +0.8246 *x where x = t-2011

YEAR	2025-26	2026-27	2027-28	2028-29	2029-30
FORECASTED	28.37	29.19	30.02	30.84	31.67

iii) For total export :-

Trend equation :Y_t =2.2941 +0.2196 *x where x = t-2011

YEAR	2025-26	2026-27	2027-28	2028-29	2029-30
FORECASTED	5.80	6.02	6.24	6.46	6.68

10 INTERPRETATION:-

1.The production and sales data are slightly positively skewed (tail to the right), indicating most years had values just below the mean.

2.Exports are nearly symmetric but show lower kurtosis, suggesting the data is relatively flat and dispersed (fewer extreme values).

3.The mean production (24.76) and sales (20.68) are substantially higher than exports (4.11), showing the domestic market's dominance.

4.Very high positive correlation (r = 0.9842) between production and domestic sales shows that as production increases, domestic sales tend to rise nearly proportionately.

5.A moderate correlation (r = 0.6688) between production and export implies export activity partially influences production.

6.The lowest correlation (r = 0.5433) is between export and domestic sales, indicating less interaction between these two markets.

7.Trend lines were fit using the least squares method, assuming a linear trend over time Irregular fluctuations (e.g., due to COVID-19) were considered outliers and excluded for the trend model.

8.Total Production Trend: Shows a steady annual increase of approximately 0.893 million units.

9.Domestic Sales Trend: Predicts an annual increase of around 0.825 million units in domestic sales, closely following production growth.

10.Export Trend: Indicates moderate growth in exports, at about 0.22 million units/year, significantly lower than production and sales.

11 CONCLUSION:-

1. **Sustained Growth:** The automobile sector in India shows **consistent linear growth** in all three dimensions—production, sales, and export.

2. **Domestic Focus:** The very high correlation between production and domestic sales suggests that the **domestic market is the primary driver** of industry growth.

3. **Moderate Export Growth:** While exports are growing, the rate is **much lower** compared to domestic sales, suggesting untapped international potential.

4. **Impact of External Shocks:** The COVID-19 period (2019–2021) caused **visible dips** in actual values, but the underlying trend remains intact, confirming **industry resilience**.

5. **Forecast Confidence:** The statistical trendlines provide a **robust forecast**, indicating the sector will continue expanding unless disrupted by major external events.

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